

Sensorimotor Rhythm or Aggregate Power? An Interpretability Audit of EEGPT on Motor Imagery

Authors [placeholder]. Affiliation [placeholder]. Track: Machine Learning and Computer Paradigms for Brain Discovery.

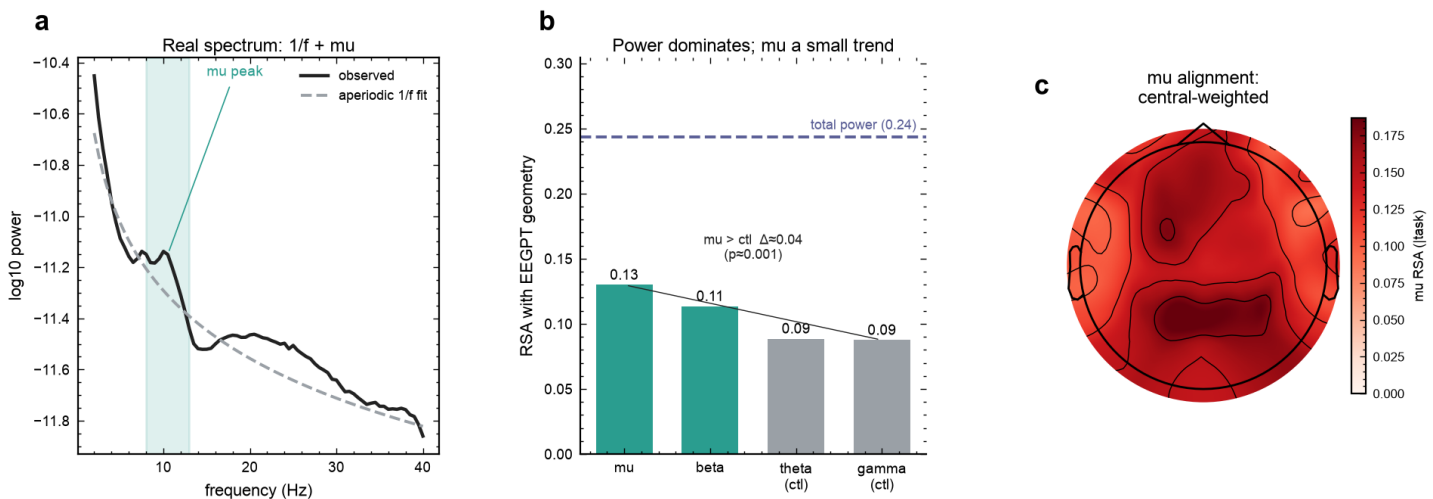
EEGPT decodes motor imagery about as well as feature engineered baselines, yet its representation is dominated by aggregate, largely aperiodic signal power. A parameterized, spatially controlled analysis finds only a small and consistent trend toward the sensorimotor mu rhythm, not a strong rhythm specific representation.

Introduction. The EEG power spectrum separates into a periodic component, the oscillations such as the sensorimotor mu and beta rhythms that index specific function [7], and an aperiodic $1/f$ component whose slope tracks excitation and inhibition balance and is itself a clinical biomarker [5]. Recent audits report that EEG foundation models lean on the aperiodic background, and the broadband power that comes with it, more than on the specific oscillations that carry task meaning [1, 2, 3, 4]. A model that reaches accuracy through aggregate power can still be useful, but it is not grounded in the rhythm a clinician would rely on. We extend this line of audits to EEGPT (NeurIPS 2024), a model absent from them and architecturally distinct: it adds a spatio-temporal representation alignment objective to masked reconstruction [6], making it a useful test of whether that architectural change alters what the model encodes. Answering it carefully matters, because raw band power conflates periodic and aperiodic activity [9] and spatial averaging can inflate apparent rhythm specificity, so the conclusion depends on doing the spectral decomposition and the spatial control properly.

Methods. We audit the frozen EEGPT encoder on PhysioNet motor imagery, imagery versus rest, across 35 subjects, using the 512 dimensional embeddings as they are used downstream. We decode imagery versus rest, and we run representational similarity analysis (RSA) [8] comparing the geometry of the embedding space to per-channel spectral references. To avoid the band power conflation, we parameterize each trial spectrum into aperiodic ($1/f$) and periodic components [5, 10] and build the oscillatory reference from aperiodic adjusted power per channel, for motor bands (mu, beta) and non-motor control bands (theta, gamma). We correlate against the embedding geometry controlling for task and for the aperiodic component, contrast central against occipital mu to separate the sensorimotor rhythm from posterior alpha, and test the mu effect against a trial shuffling null.

Results. EEGPT decodes imagery versus rest at about 71 percent within subject (chance 50), on par with feature engineered baselines, and transfers modestly across subjects. Its representational geometry is dominated by aggregate power: it correlates with total signal power at Spearman about 0.3, higher than with any single band. Once the aperiodic component is removed and the spatial construction is held fixed, only a small and consistent trend toward the sensorimotor mu rhythm remains. Aperiodic adjusted mu power is tracked about 0.04 above the non-motor control bands (mu above theta p approximately 0.002, mu above gamma p approximately 0.0003, in roughly 80 percent of subjects), it is central rather than posterior alpha (central mu survives controlling occipital, p approximately $1e-6$), and it exceeds a trial shuffling null (p approximately 0.001). Beta shows no reliable trend. We read this as a trend, not an established component: at $n = 35$ it is statistically detectable but small, and whether it persists and how precisely it can be estimated as sample size grows is an open question, since significance at a given sample size reflects detectability rather than magnitude.

Significance. EEGPT, despite its added alignment objective, represents motor imagery mainly through aggregate, largely aperiodic power, with only a small mu trend on top. This extends the aperiodic and aggregate power bias reported for reconstruction based models to a hybrid architecture, and it carries a methodological caution for the audit literature itself: whether a foundation model looks rhythm specific depends on the analysis, because raw band power conflates periodic and aperiodic activity [9] and spatial averaging inflates apparent specificity. For neurotechnology the practical reading is that EEGPT's competence rests largely on aggregate power, which is not the signal a clinician would rely on and is the part most perturbed when recording conditions shift across sessions or devices. Whether the small mu trend strengthens with more subjects, and whether representation alignment can be steered toward oscillatory rather than aperiodic structure, are open questions we are pursuing.



References

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